Sungwoo Kim

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Ph.D. Student, Computer Science, Purdue University · Anticipated graduation: May 2028 · Available Summer 2027

Education

Ph.D. in Computer Science — Purdue University *Aug 2021 – May 2028 (expected)*
West Lafayette, IN | Advisor: Dr. Dave Tian and Dongyan Xu, PurSecLab
Research: Operating Systems Security, Embedded Systems, Device Firmware, Kernel Drivers, Vulnerability Research
B.S. in Electornics and Communications Engineering — Kwangwoon University *Mar. 2014 – Feb. 2020*

Publications

- **NDSS'27**; SoK: Kicking CAN Down the Road Systematizing CAN Security Knowledge; Khaled Serag, Zhaozhou Tang, Sungwoo Kim, Vireshwar Kumar, Dave (Jing) Tian, Saman Zonouz, Raheem Beyah, Berkay Celik, and Dongyan Xu; *acceptance rate: TBD*
- **WOOT'26**; FuzzBT: Holistic-State-Guided Fuzzing for Bluetooth Host Stack in Kernels; Sungwoo Kim, Hui Peng, Imtiaz Karim, Ruoyu Wu, Jianliang Wu, Elisa Bertino, Mathias Payer, and Dave (Jing) Tian; *acceptance rate: 33%*
- **USENIX'23** Fuzz The Power: Dual-role State Guided Black-box Fuzzing for USB Power Delivery; Kyungtae Kim, Sungwoo Kim, Kevin Buttler, Antonio Bianchi, Rick Kennell, Dave (Jing) Tian; *Acceptance Rate: 29.2%*
- **ASIACCS'22**; ShadowAuth: Backward-Compatible Automatic CAN Authentication for Legacy ECUs; Sungwoo Kim, Gisu Yeo, Taegyu Kim, Junghwan "John" Rhee, Yuseok Jeon, Antonio Bianchi, Dongyan Xu, and Dave (Jing) Tian; *acceptance rate: 18.4%*

Experience

Graduate Research Assistant — PurSecLab, Purdue University *Aug. 2021 – Present*

- **Finding vulnerabilities in kernels:** Designed and built kernel fuzzers across Linux subsystems, XNU, and Android kernels, finding 19 bugs with 14 CVEs assigned.
- **Reviewing service:** Reviewer at TDSC'26, SmartSP'26, Artifact Evaluation at USENIX'24 and '26.

Vulnerability Findings

Linux Kernel
• Bluetooth: 9 CVEs [1], found via state-guided fuzz testing
• NVMe: 3 CVEs [2], found via bi-directional fuzz testing
• CXL: 1 CVE [3], found via sysfs interface fuzzing
Apple XNU kernel
• 5 Bugs; Writeup: https://sung-woo.kim/b/syz-xnu/bugs.html
Embedded systems
• Drone (PX4) : 1 CVE [4]

Technical Skills

Systems & Tool: Linux Kernel, Device Drivers, Device Controllers, Firmware, Device Tree, Clang Tooling, LLVM, Git
Board Bring-up: UART, SPI, I²C, GPIO, JTAG/SWD, Power-on Debugging
Programming Languages: C, Go, Python, JavaScript, Assembly (X86, ARM)

References

Dr. Dave Tian — Associate Professor, Purdue University (daveti@purdue.edu)

[1] CVE-2024-50255, CVE-2024-38620, CVE-2024-36968, CVE-2024-36013, CVE-2024-36012, CVE-2024-36011, CVE-2023-40283, CVE-2023-28866, CVE-2022-45934
[2] CVE-2026-23244, CVE-2026-43448, CVE-2026-43449
[3] CVE-2026-74272
[4] CVE-2021-34125